

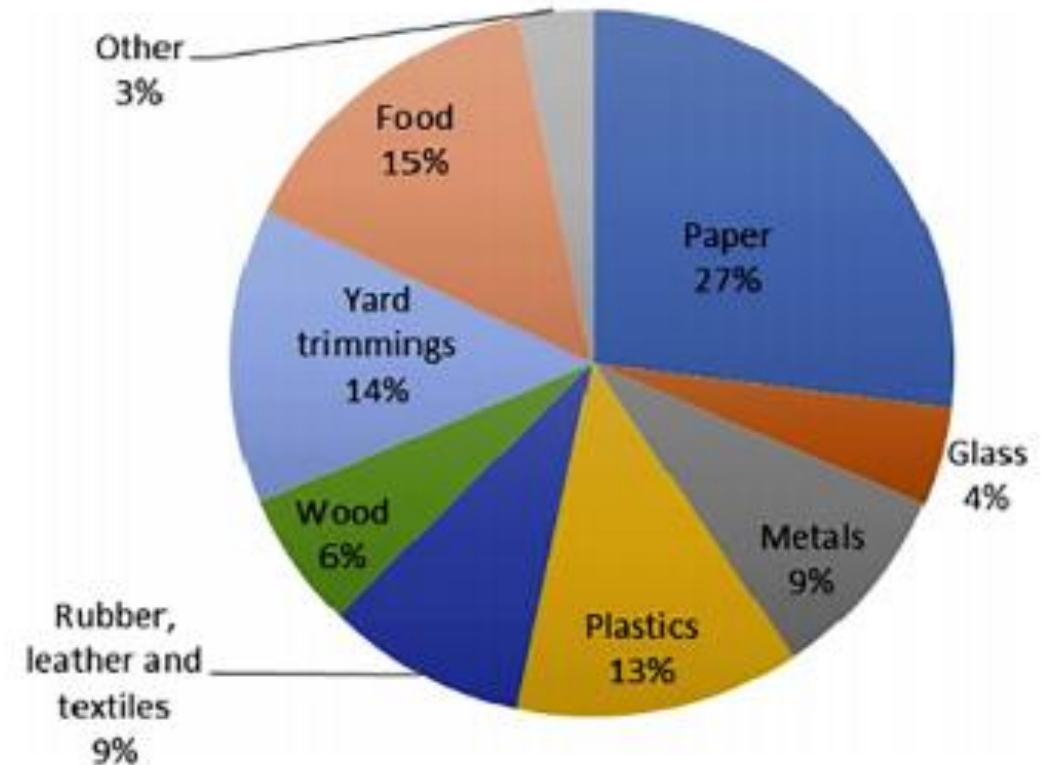
Solid Waste Management

Introduction

- Disposal of solid wastes is a stinging and widespread problem in both urban and rural areas in many developed and developing countries.
- This is mainly due to the increasing generation of such solid waste and the burden posed on the municipal budget.
- Generally, solid waste is highly; heterogeneous in nature.
- Thus, they have variable physical and chemical characteristics depending on their original sources.

Sources, composition and characterization of the solid waste

- The composition of MSW varies significantly from one municipality to another and from country to country significantly.
- Such variation depends mainly on the life style, economic situation, waste management regulations and industrial structure.
- The quantity and the composition of the municipal solid waste are critical for the determination of the appropriate handling and management of these wastes.



Types of waste:

- **Biodegradable waste:** food and kitchen waste, green waste, paper (can be recycled);
- **Recyclable material:** paper, glass bottles, cans, metals, certain plastics, etc.;
- **Inert waste:** construction and demolition waste, dirt, rocks, debris;
- **Composite wastes:** Tetra Packs, waste plastics such as toys;
- **Domestic hazardous waste & toxic waste :**
 - medication or drugs,
 - E-waste,
 - paints,
 - chemicals,
 - light bulbs or fluorescent tubes,
 - spray cans,
 - fertilizers and pesticides and their containers,
 - batteries,
 - shoe polish materials., etc.

Location	Putres -cibles	Paper	Metals	Glass	Plastics, Rubber, Leather	Tex- tiles	Ceramics, Dust, Stones	Wt (g)/ cap/ day
Bangalore, India [1]	75.2	1.5	0.1	0.2	0.9	3.1	19.0	400
Manila, Philippines [2]	45.5	14.5	4.9	2.7	8.6	1.3	27.5	400
Asunción, Paraguay [2]	60.8	12.2	2.3	4.6	4.4	2.5	13.2	460
Seoul, Korea [3]	22.3	16.2	4.1	10.6	9.6	3.8	33.4 ^a	2,000 ^a
Vienna, Austria [4]	23.3	33.6	3.7	10.4	7.0	3.1	18.9 ^b	1,180
Mexico City, Mexico [5]	59.8 ^c	11.9	1.1	3.3	3.5	0.4	20.0	680
Paris, France [4]	16.3	40.9	3.2	9.4	8.4	4.4	17.4	1,430
Australia [7]	23.6	39.1	6.6	10.2	9.9		9.0	1,870
Sunnyvale, California, USA [6]	39.4 ^d	40.8	3.5	4.4	9.6	1.0	1.3	2,000
Bexar County, Texas, USA [6]	43.8 ^d	34.0	4.3	5.5	7.5	2.0	2.9	1,816

^a Includes briquette ash (average).

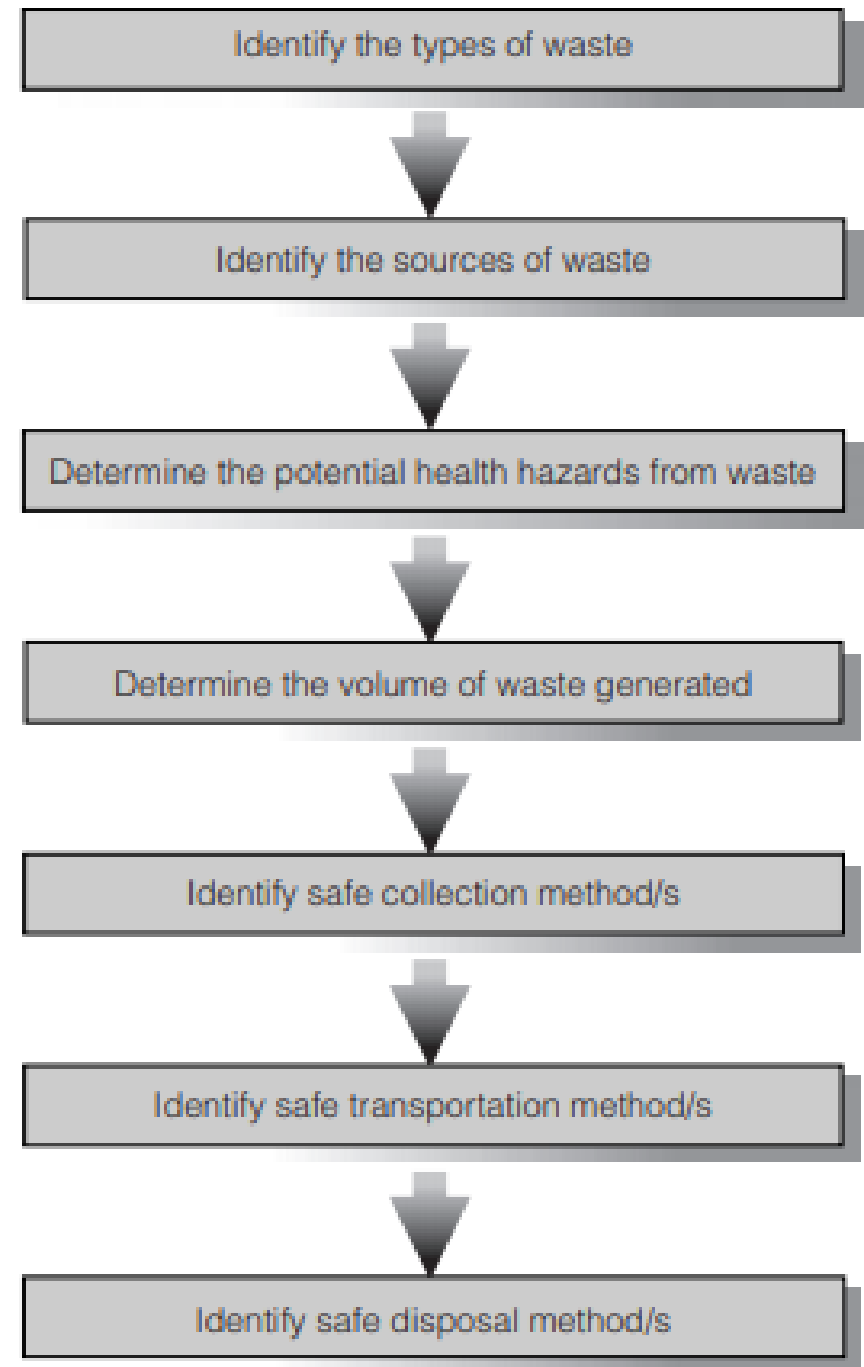
^b Includes “all others”.

^c Includes small amounts of wood, hay, and straw.

^d Includes garden waste.

Initial steps in Solid waste management and SWM hierarchy

- Prevent the production of waste, or reduce the amount generated.
- Reduce the toxicity or negative impacts of the waste that is generated.
- Reuse in their current forms the materials recovered from the waste stream.
- Recycle, compost, or recover materials for use as direct or indirect inputs to new products.
- Recover energy by incineration, anaerobic digestion, or similar processes.
- Reduce the volume of waste prior to disposal.
- Dispose of residual solid waste in an environmentally sound manner, generally in landfills.

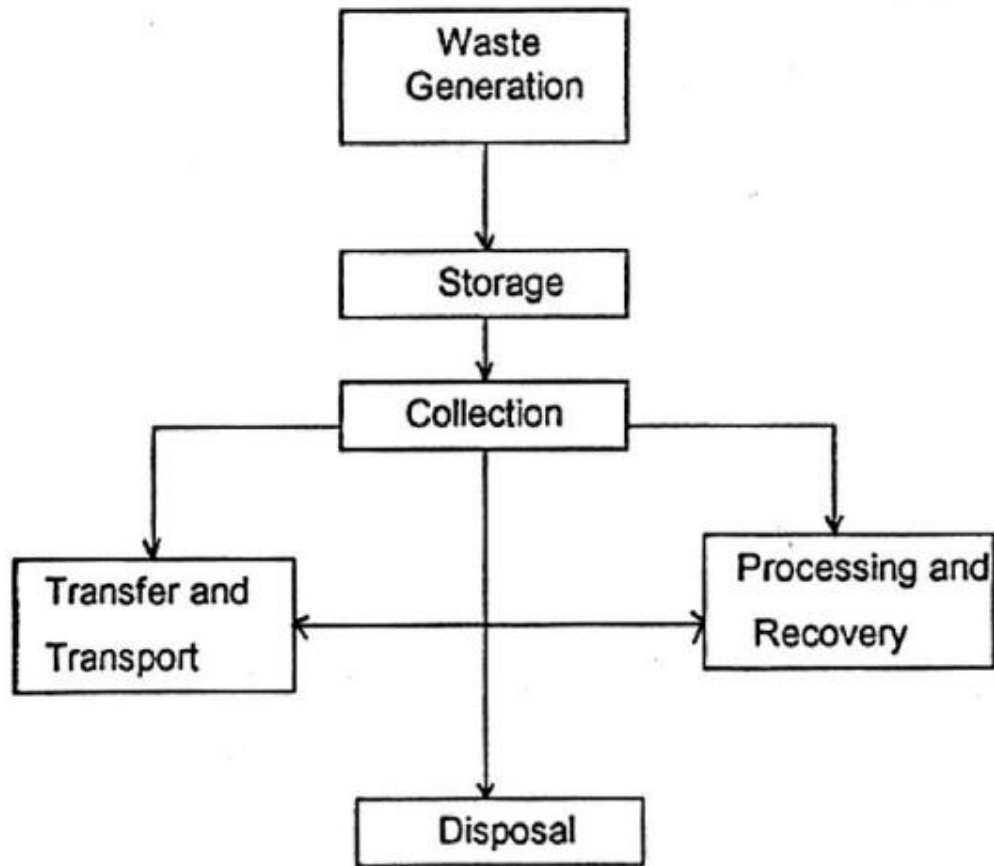


Key components of Solid waste Management

- **Collection** - the functional element of collection includes not only the gathering of solid waste and recyclable materials, but also the transport of these materials, after collection, to the location where the collection vehicle is emptied. This location may be a materials processing facility, a transfer station or a landfill disposal site.
- **Waste handling and separation, storage and processing at the source** – involves activities associated with waste management until the waste is placed in storage containers for collection. Handling also encompasses the movement of loaded containers to the point of collection. Separating different types of waste components is an important step in the handling and storage of solid waste at the source.
- **Separation and processing and transformation of solid wastes** - the types of means and facilities that are now used for the recovery of waste materials that have been separated at the source. The separation and processing of wastes that have been separated at the source and the separation of commingled wastes usually occur at a materials recovery facility, transfer stations, combustion facilities and disposal sites.

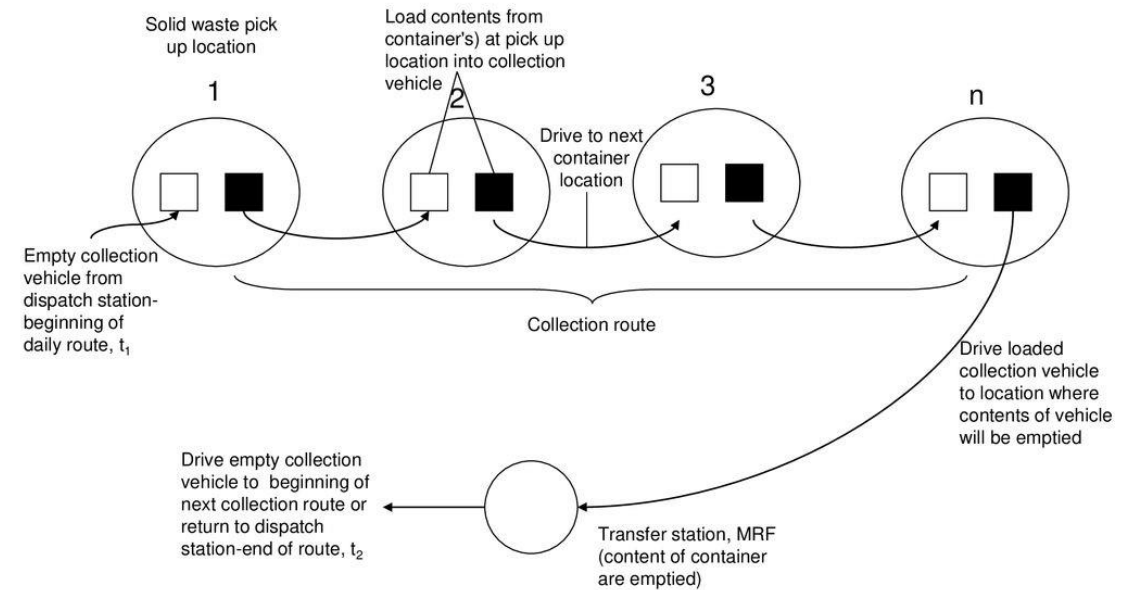
- **Transfer and transport** - this element involves two main steps. First, the waste is transferred from a smaller collection vehicle to larger transport equipment. The waste is then transported, usually over long distances, to a processing or disposal site.
- **Disposal** - The disposal of wastes by land filling or land spreading is the ultimate fate of all solid wastes, whether they are residential wastes collected and transported directly to a landfill site, residual materials from material recovery facilities, residue from the combustion of solid waste, compost, or other substances from various solid waste processing facilities.
 - A modern sanitary landfill is not a dump; it is an engineered facility used for disposing of solid wastes on land without creating nuisances or hazards to public health or safety, such as the breeding of insects and the contamination of ground water.
- **Energy generation** - municipal solid waste can be used to generate energy. Several technologies have been developed that make the processing for energy generation cleaner and more economical than ever before, including landfill gas capture, combustion, pyrolysis, gasification.

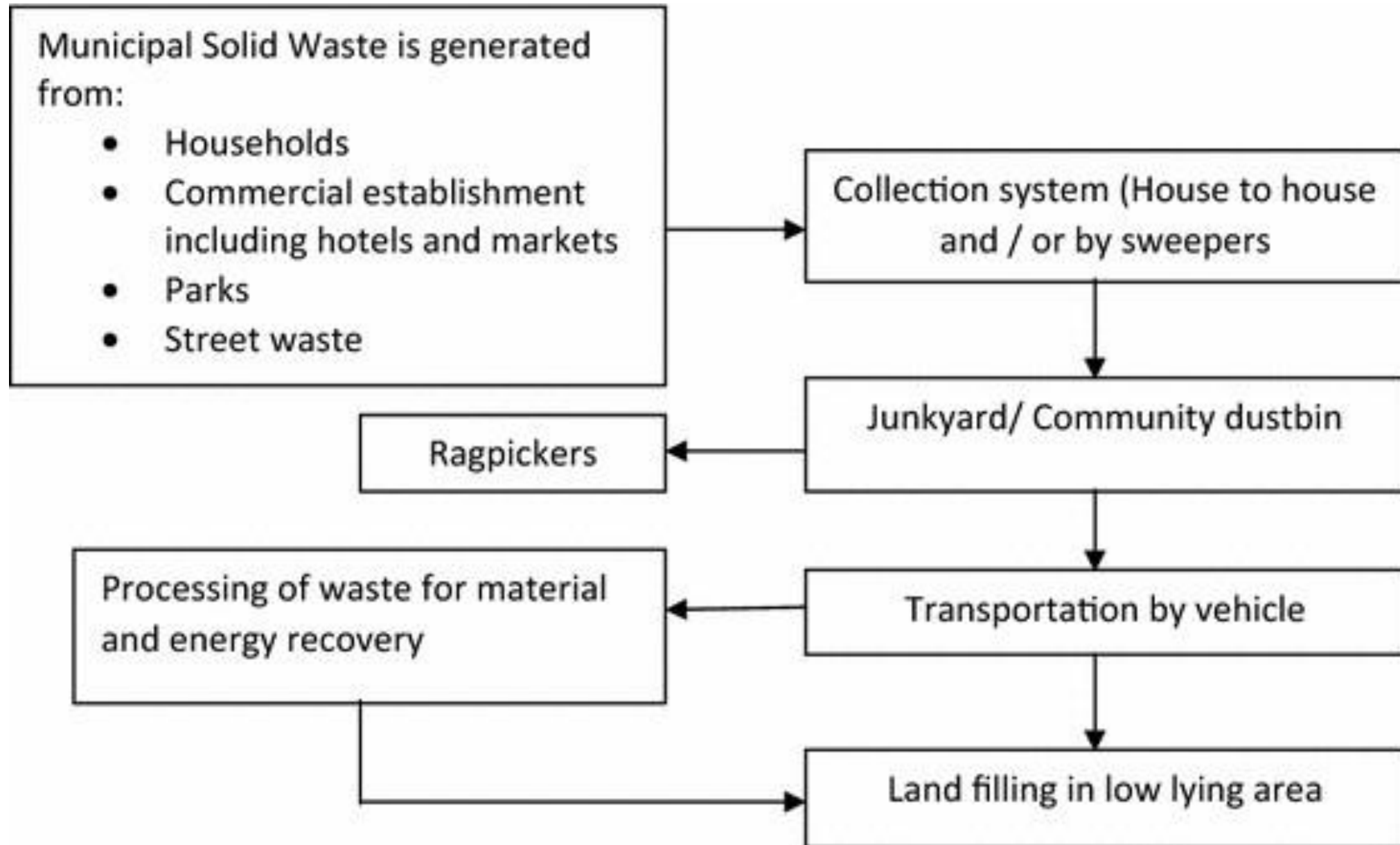
Solid waste management system elements



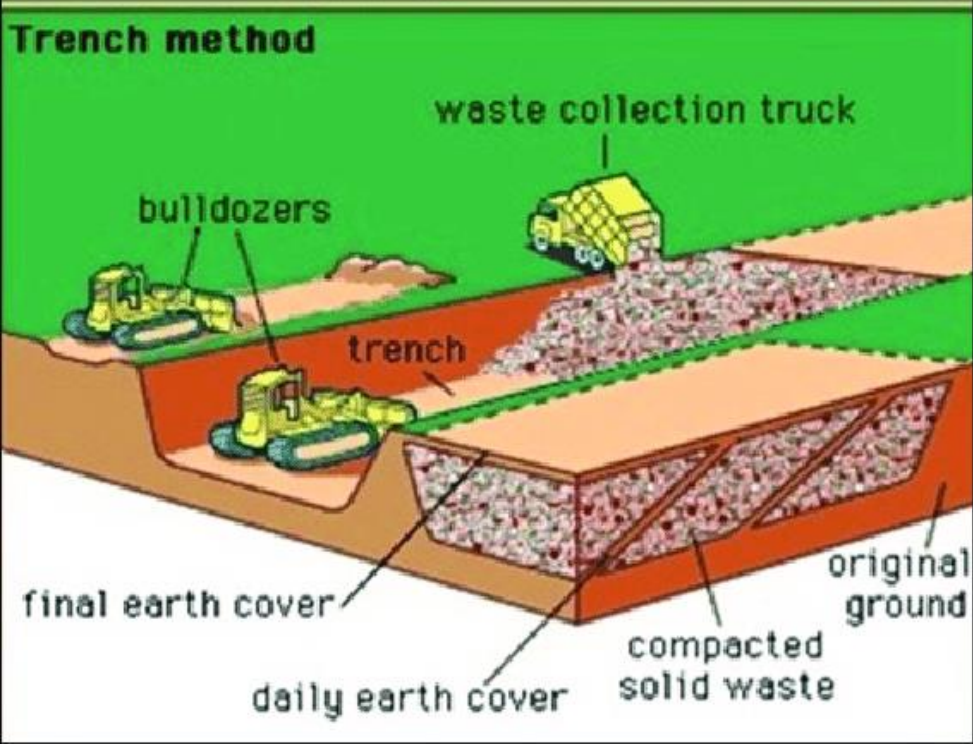
MSW COLLECTION SYSTEMS

2) STATIONARY CONTAINER SYSTEM (SCS)









Sanitary landfills

- **These are sites where waste is isolated from the environment until it is safe.** It is considered when it has completely degraded biologically, chemically and physically.
- Four basic conditions should be met before a site can be regarded as a sanitary landfill
 - Full or partial hydrogeological isolation: if a site cannot be located on land which naturally contains leachate security, additional lining materials should be brought to the site to reduce leakage from the base of the site (leachate) and help reduce contamination of groundwater and surrounding soil.



- Formal engineering preparations: designs should be developed from local geological and hydrogeological investigations. A waste disposal plan and a final restoration plan should also be developed.
- Permanent control: trained staff should be based at the landfill to supervise site preparation and construction, the depositing of waste and the regular operation and maintenance.
- Planned waste emplacement and covering: waste should be spread in layers and compacted. A small working area which is covered daily helps make the waste less accessible to pests and vermin.

Waste to energy

- There are two types of technologies used generally for converting waste streams into energy: thermal and biological. Thermal waste-to-energy is created by burning trash, while biological processes are typically centered on anaerobic digestion.
- *Energy recovery from waste is the conversion of non-recyclable waste materials into useable heat, electricity, or fuel through a variety of processes, including combustion, gasification, pyrolysis, anaerobic digestion, and landfill gas (LFG) recovery. This process is often called waste-to-energy (WTE).*

3R

- The three R's – reduce, reuse and recycle – all help to cut down on the amount of waste we throw away. They conserve natural resources, landfill space and energy. Plus, the three R's save land and money communities must use to dispose of waste in landfills. Siting a new landfill has become difficult and more expensive due to environmental regulations and public opposition.

